

Designing a High-Resolution Delta-Sigma ADC

The Quest For Precision

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1. Background & Motivation

Analog-to-Digital Converters (ADCs) are essential components in modern electronic systems, enabling the conversion of real-world analog signals into digital data for processing and storage. Among the various ADC architectures, Delta-Sigma ADCs are widely used in applications requiring high resolution and excellent noise performance, such as sensor interfaces, instrumentation systems, audio processing, and industrial measurement systems.

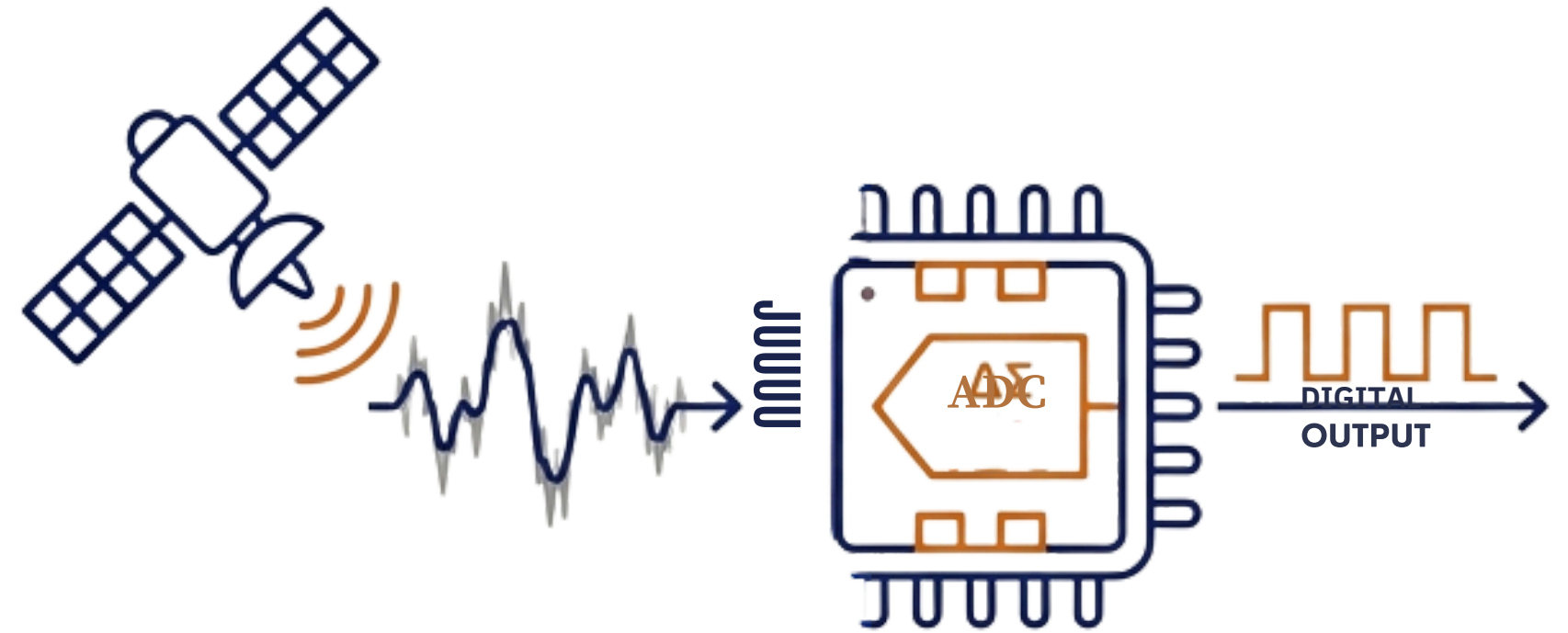
Key Advantage - Delta-Sigma ADC lies in its use of oversampling and noise shaping techniques. By sampling the input signal at a rate significantly higher than the Nyquist rate and employing a digital decimation filter, high-resolution conversion can be achieved using relatively simple analog circuitry.

This problem statement aims to introduce students to the fundamental concepts of Delta-Sigma Analog-to-Digital Conversion through the design, modelling, and simulation of a basic Delta-Sigma ADC architecture. The project focuses on understanding oversampling, quantization noise shaping, and digital decimation filtering while maintaining a manageable level of complexity suitable for educational purposes.

2. Problem Statement

The Mission: Achieving High-Precision Analog-to-Digital Conversion for Critical Space Sensors

Design, model, and simulate a basic Delta-Sigma ADC consisting of:



2.1 Delta-Sigma Modulator

Develop a first-order, single-bit Delta-Sigma Modulator comprising:

- Integrator
- 1-bit Quantizer (Comparator)
- 1-bit DAC Feedback Path

The modulator shall generate a one-bit oversampled bitstream corresponding to the analog input signal.

2.2 Digital Decimation Filter

Design and implement a digital decimation filter to recover the desired signal from the modulator output bitstream.

The filter may be implemented using: Sinc Filter, CIC (Cascaded Integrator Comb) Filter

The filter shall perform: Noise suppression, Decimation, Data rate reduction

2.3 Performance Evaluation

The complete Delta-Sigma ADC shall be simulated and evaluated using sinusoidal input signals. Performance metrics such as Signal-to-Noise Ratio (SNR) and Effective Number of Bits (ENOB) shall be calculated for different oversampling ratios.

3. Constraints

The proposed ADC shall be based on a simple first-order, single-bit Delta-Sigma Modulator followed by a digital decimation filter. The target output sampling rate shall lie between 0.5 kSPS and 2 kSPS, with oversampling ratios in the range of 64 to 256. The design should target an Effective Number of Bits (ENOB) of approximately 10–12 bits while maintaining stable operation and successful reconstruction of the input signal.

Simplifications - The study may briefly discuss advanced Delta-Sigma techniques such as higher-order modulators, multi-bit quantization, continuous-time implementations, chopping, auto-zeroing, and noise mitigation methods; however, detailed design and implementation of these techniques are outside the scope of the project. The focus shall remain on understanding the fundamentals of oversampling, noise shaping, and digital decimation.

4. Methodology

The project shall begin with a study of Delta-Sigma ADC fundamentals, including oversampling, quantization noise, and noise shaping. A first-order Delta-Sigma Modulator shall then be modeled and simulated using MATLAB/Simulink to generate an oversampled single-bit bitstream. Subsequently, a simple digital decimation filter, such as a Sinc or CIC filter, shall be designed and integrated with the modulator. The complete ADC chain shall be simulated for different oversampling ratios, and its performance shall be analyzed using frequency-domain and time-domain techniques. Finally, metrics such as SNR and ENOB shall be evaluated to study the relationship between oversampling ratio and converter performance.

5. Interface Requirements

MATLAB and Simulink shall be used for modeling, simulation, and performance analysis. The system shall accept low-frequency analog test signals, such as sinusoidal and DC inputs, and provide the Delta-Sigma bitstream, filtered digital output, spectral plots, SNR measurements, and ENOB calculations as outputs. Appropriate waveform and FFT visualizations should be included to demonstrate the operation of the modulator and decimation filter.

6. Evaluation Criteria

- Literature Study and Understanding of Delta-Sigma ADC Fundamentals – 20%
- Design and Simulation of First-Order Delta-Sigma Modulator – 25%
- Design and Implementation of Digital Decimation Filter (Sinc/CIC) – 25%
- Performance Analysis (SNR, FFT, ENOB and OSR Trade-offs) – 15%
- Technical Report and Final Presentation – 15%

7. Bonus Considerations

Additional credit may be awarded for comparative studies of different decimation filter architectures, implementation of the digital filter in Verilog/VHDL, investigation of higher-order Delta-Sigma modulators, or exploration of advanced noise reduction techniques such as chopping and auto-zeroing through simulation and literature review.